

Installing Python & VS Code

Why Proper Setup Matters

Before writing any code, it is important to set up the right tools. A correct Python and editor setup ensures a smooth learning experience and avoids common beginner issues.

Installing Python

Python is the language we will use for all data analytics tasks.

In this section, you will learn:

- How to download and install Python
- How to verify that Python is installed correctly
- Understanding the Python version and basic setup

Python is available for Windows, macOS, and Linux. Always install the **latest stable version** from the official website.

Installation

1. Open your browser and go to <https://www.python.org>
2. Click on **Downloads** and download the **latest Python version for Windows** (this will download a .exe installer).
3. Once the download is complete, **double-click the installer** to start the setup.
4. On the first setup screen, **check the box** that says **“Add Python to PATH”** (this step is very important).
5. Click on **Install Now**.
6. The installer will begin installing Python on your system. This may take a few minutes.
7. When the installation is complete, click **Close**.
8. To make sure everything works correctly, open **Command Prompt** or **Terminal** and run:
9. `python --version`
10. If Python is installed successfully, you will see the installed Python version displayed.

Your Python installation is now complete and ready to use.

Installing Visual Studio Code

Visual Studio Code is a lightweight and powerful code editor widely used by data analysts and developers.

You will learn:

- How to install VS Code
- Setting up Python support in VS Code
- Running Python files inside the editor

Installation

1. Go to <https://code.visualstudio.com>
2. Click **Download**
3. Choose your operating system
4. Download the installer
5. Run the installer
6. Accept the agreement
7. **Check these options:**
 - Add "Open with Code" action to Windows Explorer file context menu
 - Add "Open with Code" action to Windows Explorer directory context menu
 - Register Code as an editor for supported file types
 - Add to PATH
8. Click **Install**
9. Once the installation is complete, click **Finish** to launch VS Code.

And that's it! You have both Python and VS Code installed and ready to use.

Installing Python Packages using Pip

What Are Python Packages

Python packages are reusable pieces of code that add extra functionality to Python. In data analytics, packages help with tasks like data handling, calculations, and visualization.

What Is Pip

Pip is the official package manager for Python. It allows you to download and install Python packages from the Python Package Index.

Installing Packages Using Pip

To install a package, you normally use the pip command in the terminal or command prompt.

Example: `pip install package_name`

This will download and install the package so it can be used in your Python programs.

If Pip Command Does Not Work

On some systems, the pip command may not work directly due to path or environment issues. In such cases, you can use Python itself to run pip.

Use this command instead:

`python -m pip install package_name`

This ensures that pip runs using the same Python version that is installed on your system.

Verifying Installation

After installing a package, you can verify it by importing it in Python or checking the installed packages list.

By the end of this section, learners will be confident in installing and managing Python packages required for data analytics.